Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/30/2015
PI Name:	Alwood, Joshua Ph.D.		
Project Title:	Bone Loss During Simulated Weightlessness: The Role of Osteoclasts		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-1490
Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	11/01/2013	**End Date:**	03/31/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2015 (original end date was 10/31/2014) per A. Chu/ARC (Ed., 7/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Almeida, Eduardo (NASA Ames Research Center) Globus, Ruth (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts are exposed to weightlessness during spaceflight. To combat the muscle and bone loss that occurs as a result, astronauts spend a substantial amount of time exercising. Recently, zoledronate, an antiresorptive drug normally prescribed for osteoporotic patients, has been tested in astronauts (during spaceflight missions up to 6 months) with effective results of preventing bone structural losses. However, the effects of this pharmaceutical countermeasure on bone formation, damage repair, and bone quality are not fully known, especially in conditions of mechanical overload like resistance exercise. This proposal poses two principal questions: First, what are the effects of antiresorptive countermeasures combined with high-force exercise on bone remodeling and overall bone health during simulated spaceflight? Second, as pilot studies, does epigenetics play a role in the mechanical regulation of bone remodeling during simulated spaceflight? The term epigenetics describes changes in gene transcription that can affect biological adaptation to an environment--in this case, spaceflight--without changes to the sequence of DNA. Epigenetic processes are thought to underlie the process of cellular differentiation, where transcription of specific genes occurs in response to key stimuli, and may be heritable, thus, potentially affecting evolution in future generations. We hypothesize that the mechanical environment during spaceflight, namely microgravity-induced weightlessness, as well as exercise, regulate gene expression in the osteoblast-lineage cells both to control bone formation by osteoblasts and bone resorption by osteoclasts, which continually shapes bone structure throughout life. This proposal has the following Specific Aims and Hypotheses: Aim 1. Determine the effects of combined mechanical loading and zoledronate on skeletal structure and bone remodeling during simulated weightlessness. Aim 2. Demonstrate whether the methylation status of key bone remodeling gene promoters are modulated in response to various levels of mechanical loading during simulated spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project addressed multiple Research Recommendations for Animal Experiments and Gaps in Knowledge stated in the National Academy of Sciences Decadal Survey, Recapturing Space: Life and Physical Sciences for a New Era of Exploration. The knowledge generated by this research informs the high-turnover bone loss that astronauts experience while simultaneously weightless and exercising. This unique environment has similarities to Earth-bound sedentary or bed-ridden individuals being treated with bisphosphonate antiresorptives. Additionally, this work begins to investigate a fundamental molecular mechanism, specifically, DNA methylation, that may improve our fundamental understanding of skeletal regulation and the organism's response to spaceflight.
Task Progress:	In our first project, we developed an integrated model that simulates the combination of weightlessness, exercise-induced mechanical strain, and antiresorptive treatment (zoledronate) that astronauts may experience during space missions. We observed a predictable effect from each independent variable. Subtle interactions were observed, though none are predicted to affect structural integrity. Ongoing work is investigating whether combined treatments affected in situ cellular activity and structural integrity. Altogether, our results suggest that, at the cancellous-tissue and cortical level, zoledronate treatment or tibial compression during simulated spaceflight was protective of the negative insults of disuse. In our second project, we queried whether DNA methylation played a role in the skeleton's response to mechanical loading or musculoskeletal disuse. Our studies characterized a low level of methylation present in the tibia and at promoters for key genes related to bone cells. Further work is needed to elucidate whether simulated spaceflight or exercise affects DNA methylation.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Stevick RJ, Tran LH, Nalavadi M, Almeida EAC, Globus RK, Alwood JS. "Does simulated weightlessness alter the methylation status of gene promoters during bone remodeling?" Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Nalavadi MO, Shirazi-Fard Y, Castillo AB, Alwood JS. "The effects of combined zolendronate treatment and mechanical loading during simulated weightlessness on bone structure." Presented at the 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Awards	Alwood JS. "2012 Presidential Early Career Award for Scientists and Engineers, December 2013." Dec-2013